Robust Nonlocal Low-Rank SAR Time Series Despeckling Considering Speckle Correlation by Total Variation Regularization

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***Abstract*— Outliers and speckle both corrupt time series of synthetic aperture radar (SAR) acquisitions. Owing to the coherence between SAR acquisitions, their speckle can no longer be regarded as independent. In this study, we propose an algorithm for nonlocal low-rank time series despeckling, which is robust against outliers and also specifically addresses speckle correlation between acquisitions. By imposing total variation regularization on the signal's speckle component, the correlation between acquisitions can be identified, facilitating the extraction of outliers from unfiltered signals and the correlated speckle. This robustness against outliers also addresses matching errors and inaccuracies in the nonlocal similarity search. Such errors include mismatched data in the nonlocal estimation process, which degrade the denoising performance of conventional similarity-based filtering approaches. Multiple experiments on real and synthetic data assess the performance of the approach by comparing it with state-of-the-art methods. It provides filtering results of comparable quality but is not adversely affected by outliers. The source code is available at https://github.com/gbaier/nllrtv.**

***Index Terms*— Despeckling, low-rank, multitemporal, nonlocal filtering, synthetic aperture radar (SAR), total variation (TV).**

I. Introduction

With the launch of the European space agency's Sentinel-1 satellites [1], synthetic aperture radar (SAR) stacks have become commonplace, owing to the high coverage frequency and wide swath widths of these satellites. Future satellite missions, such as ALOS-4 [2], NISAR [3], and Tandem-L [4], have comparable or superior acquisition capabilities. These systems are capable of acquiring the entire globe within one or two weeks, allowing detailed analyses of even quickly varying, dynamic processes. Deforestation, crop monitoring, urbanization, and infrastructure monitoring are some examples that show the usefulness and impact of frequent and large-scale acquisitions.

However, SAR images are affected by speckle, inherent to coherent acquisition systems, rendering any further analysis more challenging. Despeckling of single SAR images has been an active research topic for more than three decades; [5] provides a comprehensive review. Yet, taking advantage of the temporal domain when filtering SAR stacks has only recently received more attention. Aside from being corrupted by speckle, SAR time series additionally exhibit temporal changes and outliers, thus complicating the filtering process. These changes and outliers can be weather effects, seasonal changes, radio frequency interference, or sporadically appearing point scatterers, such as ships or other vehicles.

In the past, multitemporal SAR despeckling filters were direct adaptations of existing spatial approaches [6]–[8]. Either by using the temporal neighborhood as filter input instead of the spatial neighborhood, i.e., for computing weighted averages, or extending the spatial search area with the temporal axis, i.e., for adaptive neighborhood filters. A different approach is taken by [9], which decomposes the SAR stack into a slowly varying background signal, strong scatterers, and a noise component, making use of their different temporal characteristics.

Many advanced stack filtering algorithms rely on the nonlocal filtering paradigm [10], which exploits the fact that natural scenes exhibit repeating patterns and structures, often in unconnected neighborhoods. By identifying and jointly filtering them, impressive denoising performances are achieved. Although originally applied for denoising regular optical images, nonlocal filters were later adapted to and have shown their worth for SAR despeckling [11]–[14], polarimetric SAR [15]–[17], SAR interferometry [18]–[22], and general SAR covariance matrix estimation [23], applicable to all of the abovementioned domains.

In [24], the authors proposed a nonlocal filter for SAR time series. The method first computes a guidance image [25] by filtering in the temporal domain, which then helps find similar structures in the second step more accurately. As another example, [26] extends SAR-BM3D [12] from single images to time series. It also searches for similar patches nonlocally and reduces the noise level by a collective wavelet transform and subsequent thresholding.

Other methods apply transforms so that the SAR noise characteristics correspond to standard, additive Gaussian noise [27], [28]. By combining these methods with state-of-the-art nonlocal filters for Gaussian noise [29], comparable performance with dedicatedly developed multitemporal despeckling filters can be achieved.

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In this article, we propose a method that combines the nonlocal filtering paradigm with low-rank approximation, total variation (TV) regularization, and sparse outlier removal. This nonlocal low-rank denoising can tackle speckle was already demonstrated in [30] and [31] for single images. Our motivation for an extension to stacks of SAR images is twofold. Low-rank methods can inherently deal with temporal outliers that would otherwise hamper further processing. Another problem is the nonlocal filtering process itself. Nonlocal filters group similar patches in a neighborhood as one filtering step. If this grouping is imperfect, which is more likely for large patches that naturally occur with stacks, it introduces bias to the estimation process. A method that identifies these artifacts as sparse outliers and removes them avoids corrupting the filtering result and eases any subsequent analysis. The role of the TV regularization is to impose a certain regularity on the group's elements, both in temporal and spatial directions, which further improves the filtering quality.

The remainder of this article is structured as follows. In Section II, we present the nonlocal filtering paradigm, and introduce low-rank approximation of matrices and the benefits of combining both. We present an extension to nonlocal low-rank approximation to account for the temporal speckle correlation in Section III, together with an exemplary SAR stack despeckling filter that makes use of this extension. In Section IV experiments on simulated and real-world data compare the proposed method with state-of-the-art algorithms. These are finally followed by a discussion of the results and concluding remarks in Sections V and VI. The source code is available at https://github.com/gbaier/nllrtv.

II. Nonlocal Low-Rank Denoising

This section provides a brief introduction to the two most important underlying techniques of the proposed method.

A. Low-Rank Approximation

In recent years, low-rank matrix approximation has seen increasing popularity, starting with the seminal work in [32] and [33] (first published in [34] and [35]). The authors showed that, under mild constraints, a low-rank matrix $\mathbf{B}$ corrupted by sparse outliers $\mathbf{S}$ can be decomposed into both components by minimizing

$$\begin{aligned} &\arg\min_{\mathbf{B},\mathbf{S}} \|\mathbf{B}\|_* + \lambda\|\mathbf{S}\|_1 \\ &\text{s. t.} \quad \mathbf{G} = \mathbf{B} + \mathbf{S} \end{aligned} \tag{1}$$

where $\|\mathbf{B}\|_* = \sum_i \sigma_i(\mathbf{B})$ is the sum over all singular values $\sigma_i(\mathbf{B})$ of $\mathbf{B}$ (i.e., the nuclear norm), $\|\mathbf{S}\|_1 = \sum_{ij} \mathbf{S}_{ij}$ is an L_1 penalization and $\mathbf{G}$ is the corrupted measurements. Noting the similarity of (1) to the principal component analysis (PCA) minimization problem [32] also referred to their approach as robust PCA. Remarkably, a fixed formula for λ leads to convergence with almost certainty, avoiding an onerous parameter search. Furthermore, they proposed a simple, yet efficient algorithm for solving the minimization problem.

Later, [36] extended the model in (1) to also deal with additive Gaussian noise in addition to sparse outliers; that is, $\mathbf{G} = \mathbf{B} + \mathbf{S} + \mathbf{N}$, with $\mathbf{N}$ being a dense matrix, the entries of which are normally distributed with a small variance compared with the outliers in $\mathbf{S}$. Another refinement was presented in [37] and [38], which showed it to be advantageous to modify the simple nuclear norm by assigning each singular value in the nuclear norm a corresponding weight w_i

$$\|\mathbf{B}\|_{\mathbf{w}*} = \sum w_i \sigma_i(\mathbf{B}). \tag{2}$$

Downweighting large singular values leads to lower ranks of $\mathbf{B}$, which better reflects signals encountered in nature, resulting in better low-rank approximations.

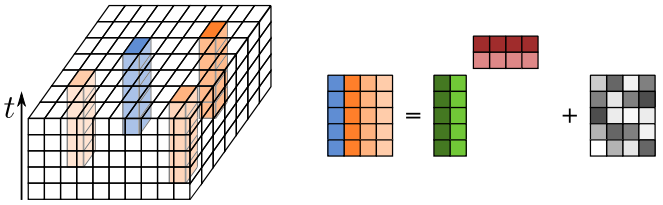


Fig. 1. Illustration of nonlocal low-rank denoising. Similar vectors are stacked along a new axis to form a matrix. This matrix exhibits low-rank, although perturbed by noise, since all the column vectors stem from a similar process and can be decomposed into both parts.

B. Nonlocal Self-Similarity

The first proposed nonlocal filter [10] was essentially a generalization of the bilateral filter [39]. For each pixel of the image, it also identifies similar pixels and computes the denoised output as a weighted mean. However, it differs in two critical aspects from the bilateral filter.

1) Its search for similar pixels compares not only just the target and the candidate pixels but also their surrounding patches. This way, textures and features help identifying similar pixels and more accurate and robust weights are computed.
2) The distances of the candidate pixels to the target pixel do not affect the weights. However, mostly due to computational constraints, the search for similar pixels is usually restricted to a comparatively large window.

The original nonlocal filter's performance sparked research resulting in several refinements. Instead of the simple weighted mean, more advanced methods generally take the collection of similar patches and exploit their similarity. Notable approaches are: collective wavelet transform and thresholding [29], group sparse approximation using a dictionary [40], and low-rank approximation [41].

Section III covers the rather straightforward extension of nonlocal low-rank approximation to collections of images and also the rationale why the simple nonlocal means can fail in such a setting.

C. Putting Both Together

Fig. 1 illustrates the filtering process for a set of five images, stacked along the temporal axis t to form a 3-D tensor. Similar time series vectors are identified and a matrix is constructed by stacking them along a new axis. As this matrix is formed by

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similar vectors, a reasonable assumption is that without perturbing noise, it would exhibit low rank. Solving (1) separates both components, and the low-rank approximation is written to the column vectors' original positions. The process is repeated for each spatial coordinate, while aggregating estimates with the same coordinates.

This procedure would obviously also apply to a single image by replacing vectors with patches, which are flattened to form columns of a matrix for low-rank approximation [31], [41].

Compared with the traditional nonlocal means, the nonlocal low-rank approximation provides benefits in terms of robustness, which we will detail in the following paragraphs.

We have already established that one advantage of the nonlocal filters over the bilateral filter is their use of patches instead of single pixels for computing similarities between the pixels. This is usually done by taking the mean over all contained pixel similarities between two patches. Although this gives a more accurate estimate of the similarity, it also naturally decreases the influence of single pixels on the similarity estimate. This increases the probability to include outliers into the estimation process, given that their contribution to the similarity computation might be outweighed by other pixels in the patch. For denoising time series, as sketched out in Fig. 1, this becomes even more relevant since, with more images in a stack, the patch size also increases.

Notably for SAR image stacks, a similar problem arises for isolated point scatterers. If their distinctly different brightness is not analogously reflected in the calculated similarity, a nonlocal means filter will smooth them out, simultaneously smudging their surroundings.

A low-rank approximation addresses both issues by removing outliers from the estimation process and preserving the structure contained in the collection of vectors to a larger degree than the simple weighted average in conventional nonlocal means filters. Thus, mistakes that are made when matching similar amplitude vectors will not be as critical as with traditional nonlocal means filters.

III. Nonlocal Low-Rank SAR Stack Denoising

In this section, we extend (1) to account for certain characteristics of SAR stacks and present an algorithm to solve the resulting optimization problem.

A. Consideration of Speckle Correlation in SAR Stacks

We start by showing how the SAR signal can be cast into a model of the form (1). A common SAR amplitude signal model is [5]

$$g = \beta u \tag{3}$$

where g is the observed image, β is the true radar brightness (i.e., the noise free signal), and u is an exponentially distributed random variable of mean 1, modeling speckle. This model can trivially be transformed into

$$g = \beta + (u-1)\beta \tag{4}$$

with an additive, signal-dependent noise term $(u-1)\beta$ describing the speckle "noise." By going from a single pixel g to a collection of time series, stacked along a new axis as a matrix $\mathbf{G}$, speckle and outlier removal can be cast as a low-rank approximation

$$\begin{aligned} &\underset{\mathbf{B},\mathbf{N}+\mathbf{S}}{\arg\min} \ \|\mathbf{B}\|_{\mathbf{w}*} + \lambda\|\mathbf{N}+\mathbf{S}\|_1 \\ &\text{s.t. } \mathbf{G} = \mathbf{B}+\mathbf{N}+\mathbf{S} \end{aligned} \tag{5}$$

with a straightforward extension of (1) by the weighted nuclear norm [37], [38]. The problem bears some similarity to the TV-L_1 problem [42] for image denoising, with the exception that the TV regularization is replaced with a low-rank penalization. Enforcing low rank on the collection of similar time series $\mathbf{B}$ ensures a certain regularity and smoothness. Exactly due to their similarity, assuming low rank of $\mathbf{B}$ is well-founded. Conversely, $\mathbf{N}+\mathbf{S}$, captures dense, small amplitude speckle noise $\mathbf{N}$ and sparse, large amplitude outliers $\mathbf{S}$ in the nonlocal grouping. This is reminiscent of the earlier mentioned work [36], which performs a low-rank approximation in the presence of outliers and Gaussian noise. A similar minimization problem was also used in [30] for ultrasound speckle removal, empirically demonstrating its applicability to dense noise. Note that using (5), we are unable to separate speckle noise from outliers. In order to do so, we can take advantage of certain characteristics of the SAR speckle in time series.

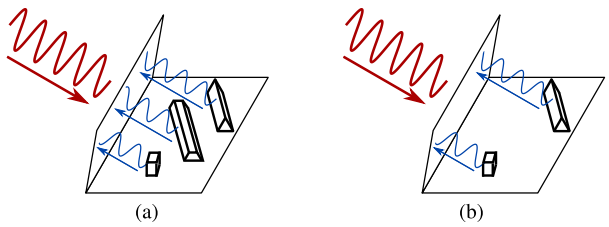


Fig. 2. Origin of speckle and speckle correlation between two acquisitions [(a) and (b)]. Inside one SAR resolution cell, multiple elemental scatterers reflect the impinging electromagnetic wave. Due to their constructive and destructive interference, the total received signal's amplitude at the SAR sensor fluctuates from one pixel to another, so-called speckle. In case the composition of elemental scatterers changes, the received signal's speckle component also changes, although it remains correlated with a certain degree depending on the coherence.

Inside one SAR resolution cell, multiple elemental scatterers reflect the impinging electromagnetic wave, as shown in Fig. 2(a). The superposition of all reflections accounts for the received signal at the SAR sensor. Due to their different phases, the individual components interfere constructively and destructively, making the total received amplitude a random process. As the composition of elemental scatterers changes between adjacent resolution cells, regardless of whether the underlying parameters such as surface roughness, soil moisture, and so on are identical, visible signal fluctuations can occur even in homogeneous areas. This deviation from the true radar backscatter is referred to as speckle.

A common assumption is that all elemental scatterers inside a resolution cell have approximately the same backscatter; that is, their return signals have identical amplitude and their number tends to infinity. Furthermore, if the surface is assumed to be rough with respect to the wavelength, the various reflected waves will have a uniformly distributed phase

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between $-\pi$ and π. Under these conditions, the total received signal follows a complex Gaussian random distribution, with a Rayleigh distributed amplitude and a uniformly distributed phase [43]. This is generally referred to as *fully developed* speckle.

If between two SAR acquisitions the composition of elemental scatterers changes [see Fig. 2(b)], the total received signal at the SAR sensor is also altered, while still retaining parts of the original signal. In other words, although speckle is not correlated in space, it is correlated in time. Guarnieri and Prati [44] exploit this characteristic for an amplitude-based coherence estimator. Another example that takes advantage of correlated speckle is speckle tracking [45], which estimates the motion of glaciers.

We tried to capture this characteristic by decomposing the signal into three components as opposed to the two-component decomposition in (5): a low-rank signal component $\mathbf{B}$, temporarily correlated speckle $\mathbf{N}$, and uncorrelated speckle plus outliers $\mathbf{S}$. The temporarily correlated speckle is imposed by further regularizing $\mathbf{N}$ using TV [46] along the temporal dimension

$$\begin{aligned}&\underset{\mathbf{B},\mathbf{N},\mathbf{S}}{\arg\min}\|\mathbf{B}\|_{\mathbf{w}*}+\mu\|\mathbf{N}\|_1+\nu\sum|\nabla_t\mathbf{N}|+\|\mathbf{S}\|_1\\&\text{s. t. }\mathbf{G}=\mathbf{B}+\mathbf{N}+\mathbf{S}.\end{aligned}\tag{6}$$

The benefit of this three-component decomposition is the separation of outliers from speckle, at least the temporarily coherent part. This facilitates their further detection and analysis, as shown in Section IV.

The observant reader might question the choice of the L_1 norm to regularize $\mathbf{N}$ instead of the more widely used L_2 norm. The reasons are as follows.

1) L_2 particularly penalizes perturbations of large magnitude. As speckle is amplitude-dependent, i.e., increases for brighter areas, this might lead to $\mathbf{S}$ and $\mathbf{B}$ absorbing some of the speckle.
2) The decomposition of noise into speckle and outliers is an intricate problem, requiring a careful choice of regularization parameters. For the large backscatter variations of an SAR image, reasoning and finding such a parameter set are much easier if both speckle and noise are similarly regularized. We specifically address the choice of parameters in Section III-C.

As a final refinement, in our experiments, we found that additionally imposing TV, in this case in both temporal and spatial dimensions, on the low-rank component $\mathbf{B}$ further improves the quality of the estimate, similar to the approach proposed in [47] for hyperspectral image denoising. The rationale for smoothness is different for both dimensions. We can assume that along the temporal axis changes happen gradually, whereas along the spatial axis, the overall similar time series should have comparable radar backscatter. This leads to the following final optimization problem:

$$\begin{aligned}&\underset{\mathbf{B},\mathbf{N},\mathbf{S}}{\arg\min}\|\mathbf{B}\|_{\mathbf{w}*}+\lambda\sum|\nabla\mathbf{B}|+\mu\|\mathbf{N}\|_1+\nu\sum|\nabla_t\mathbf{N}|+\|\mathbf{S}\|_1\\&\text{s.t. }\mathbf{G}=\mathbf{B}+\mathbf{N}+\mathbf{S}.\end{aligned}\tag{7}$$

We would also like to point out some similarities to the nonlocal TV approach introduced in [48], which computes derivatives on a nonlocal, weighted graph. To some degree, this resembles our approach, which collects nonlocal samples and then enforces regular TV. Both approaches assume that it is beneficial to enforce smoothness not only on neighboring pixels but also on nonadjacent, but similar pixels.

B. Optimization Problem

Equation (7) can be solved by extending the algorithm for solving the weighted nuclear-norm robust PCA problem proposed in [38], which employs the method of augmented Lagrange multiplier (ALM) [49], [50].

Introducing the latent variable $\mathbf{T}=\mathbf{B}$

$$\begin{aligned}&\underset{\mathbf{B},\mathbf{N},\mathbf{S},\mathbf{T}}{\arg\min}\ \|\mathbf{B}\|_{\mathbf{w}*}+\lambda\sum|\nabla\mathbf{T}|+\mu\|\mathbf{N}\|_1+\nu\sum|\nabla_t\mathbf{N}|+\|\mathbf{S}\|_1\\&\text{s.t. }\mathbf{G}=\mathbf{B}+\mathbf{N}+\mathbf{S}\text{ and }\mathbf{B}=\mathbf{T}\end{aligned}\tag{8}$$

allows us to tackle the low-rank and TV regularizations on $\mathbf{B}$ separately. The corresponding Lagrangian of (8) is

$$\begin{aligned}&\mathcal{L}(\mathbf{B},\mathbf{N},\mathbf{S},\mathbf{T},\mathbf{\Lambda}_1,\mathbf{\Lambda}_2)\\&=\|\mathbf{B}\|_{\mathbf{w}*}+\lambda\sum|\nabla\mathbf{T}|+\mu\|\mathbf{N}\|_1+\nu\sum|\nabla_t\mathbf{N}|+\|\mathbf{S}\|_1\\&\quad+\langle\mathbf{\Lambda}_1,\mathbf{G}-\mathbf{B}-\mathbf{N}-\mathbf{S}\rangle+\langle\mathbf{\Lambda}_2,\mathbf{T}-\mathbf{B}\rangle\\&\quad+\frac{\alpha}{2}\|\mathbf{G}-\mathbf{B}-\mathbf{N}-\mathbf{S}\|_{\mathrm{F}}^2+\frac{\alpha}{2}\|\mathbf{T}-\mathbf{B}\|_{\mathrm{F}}^2\end{aligned}\tag{9}$$

where $\mathbf{\Lambda}_1$ and $\mathbf{\Lambda}_2$ denote the Lagrangian multipliers. ALM iteratively and separately solves the four subproblems for $\mathbf{B}$, $\mathbf{N}$, $\mathbf{S}$, and $\mathbf{T}$. Algorithm 1 shows the corresponding pseudocode.

Algorithm 1 ALM for Solving (8). The Guidelines for Parameter Selection Are Explained in Section III-C

Input : data matrix $\mathbf{G}\in\mathbb{R}^{m_1\times m_2}$, low-rank penalty C, low-rank TV penalty λ, speckle L_1 penalty μ, speckle TV penalty ν
Output: low-rank $\mathbf{B}$, outliers $\mathbf{S}$

$\mathbf{B}_0=\mathbf{G}$
$\mathbf{N}_0=\mathbf{S}_0=\mathbf{T}_0=\mathbf{\Lambda}_{1,0}=\mathbf{\Lambda}_{2,0}=\mathbf{0}$
$\alpha=\frac{m_1\ m_2}{4\|\mathbf{G}\|_1}$
$\mathbf{U}\Sigma\mathbf{V}^\dagger=\mathrm{SVD}(\mathbf{G})$
$w_i=C/(\sqrt{\Sigma_{i,i}}+\varepsilon)$
while *not converged* **do**
  $\mathbf{S}_{k+1}=\mathcal{S}_{\frac{1}{\alpha}}\left(\mathbf{G}+\mathbf{\Lambda}_{1,k}/\alpha-\mathbf{B}_k-\mathbf{N}_k\right)$
  `// SVD soft-thresholding`
  $\mathbf{U}\Sigma\mathbf{V}^\dagger=\mathrm{SVD}\left(\mathbf{G}+\mathbf{T}_k+\frac{\mathbf{\Lambda}_{1,k}+\mathbf{\Lambda}_{2,k}}{\alpha}-\mathbf{N}_k-\mathbf{S}_{k+1}\right)$
  $\mathbf{B}_{k+1}=\mathbf{U}\mathcal{S}_{\frac{w}{2\alpha}}\left(\frac{1}{2}\Sigma\right)\mathbf{V}^\dagger$
  $\mathbf{T}_{k+1}=\mathrm{TV}_{\frac{\lambda}{\alpha}}\left(\mathbf{B}_{k+1}-\frac{\mathbf{\Lambda}_{2,k}}{\alpha}\right)$
  $\mathbf{N}_{k+1}=\text{TV-L1}_{\nu,\alpha}\left(\mathbf{G}+\frac{\mathbf{\Lambda}_1}{\alpha}-\mathbf{B}_{k+1}-\mathbf{S}_{k+1}\right)$
  `// Update Lagrangians`
  $\mathbf{\Lambda}_{1,k+1}=\mathbf{\Lambda}_{1,k}+\alpha\left(\mathbf{G}-(\mathbf{B}_{k+1}+\mathbf{N}_{k+1}+\mathbf{S}_{k+1})\right)$
  $\mathbf{\Lambda}_{2,k+1}=\mathbf{\Lambda}_{2,k}+\alpha\left(\mathbf{T}_{k+1}-\mathbf{B}_{k+1}\right)$
  $k=k+1$
end

The following paragraphs describe in detail how to solve the various subproblems. Guidelines for selecting parameters

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in (9) follow. The solutions for two of the subproblems make use of the soft thresholding operator defined as

$$\mathcal{S}_\lambda(x) \ := \text{sign}(x)\ (|x| - \lambda)_+.$$

1) Sparse Outliers and Uncorrelated Noise **S***:* There exists a closed-form solution to

$$\begin{aligned}\mathbf{S}_{k+1} &= \arg\min_{\mathbf{S}} \mathcal{L}(\mathbf{B}_k, \mathbf{N}_k, \mathbf{S}, \mathbf{T}_k, \mathbf{\Lambda}_{1,k}, \mathbf{\Lambda}_{2,k}) \\ &= \arg\min_{\mathbf{S}} \|\mathbf{S}\|_1 + \frac{\alpha}{2}\left\|\mathbf{G} + \frac{\mathbf{\Lambda}_{1,k}}{\alpha} - \mathbf{B}_k - \mathbf{N}_k - \mathbf{S}\right\|_{\text{F}}^2 \end{aligned} \tag{10}$$

in the form of simple soft thresholding

$$\mathbf{S}_{k+1} = \mathcal{S}_{\frac{1}{\alpha}}\left(\mathbf{G} + \frac{\mathbf{\Lambda}_{1,k}}{\alpha} - \mathbf{B}_k - \mathbf{N}_k\right). \tag{11}$$

2) Low-Rank Component **B***:* The solution for

$$\begin{aligned}\mathbf{B}_{k+1} &= \arg\min_{\mathbf{B}} \mathcal{L}(\mathbf{B}, \mathbf{N}_k, \mathbf{S}_{k+1}, \mathbf{T}_k, \mathbf{\Lambda}_{1,k}, \mathbf{\Lambda}_{2,k}) \\ &= \arg\min_{\mathbf{B}} \|\mathbf{B}\|_{\mathbf{w}*} + \langle \mathbf{\Lambda}_{1,k}, \mathbf{G} - \mathbf{B} - \mathbf{N}_k - \mathbf{S}_{k+1}\rangle \\ &\quad + \langle \mathbf{\Lambda}_2, \mathbf{T}_k - \mathbf{B}\rangle + \frac{\alpha}{2}\|\mathbf{G} - \mathbf{B} - \mathbf{N}_k - \mathbf{S}_{k+1}\|_{\text{F}}^2 \\ &\quad + \frac{\alpha}{2}\|\mathbf{T}_k - \mathbf{B}\|_{\text{F}}^2 \\ &= \arg\min_{\mathbf{B}} \|\mathbf{B}\|_{\mathbf{w}*} + \frac{\alpha}{2}\|\mathbf{B} \\ &\quad - \frac{1}{2}\left(\mathbf{G} + \mathbf{T}_k + \frac{\mathbf{\Lambda}_{1,k} + \mathbf{\Lambda}_{2,k}}{\alpha} - \mathbf{N}_k - \mathbf{S}_{k+1}\right)\Bigg\|_{\text{F}}^2 \end{aligned} \tag{12}$$

is of the form

$$\arg\min_{\mathbf{B}} \|\mathbf{B}\|_{\mathbf{w}*} + \|\mathbf{B} - \mathbf{Y}\|_{\text{F}}^2. \tag{13}$$

Let the singular values be sorted in descending order in the weighted nuclear norm definition [see (2)]. In case the weights are ordered in nondescending order, i.e., we match large singular values with small weights to promote low rank, there exists a closed-form solution to (13) [38]. Under these conditions

$$\mathbf{B}_{k+1} = \mathbf{U}\mathcal{S}_{\frac{\mathbf{w}}{2}}(\Sigma)\,\mathbf{V}^*$$

where $\mathbf{Y} = \mathbf{U}\Sigma\mathbf{V}^*$ is the singular value decomposition of $\mathbf{Y}$ and $\mathcal{S}_{\mathbf{w}/2}$ is the generalized soft-thresholding operator with weighting vector $\mathbf{w}$, which thresholds $\Sigma_{i,i}$ with $\mathbf{w}_i$, solves the weighted nuclear norm minimization problem.

3) TV on Low-Rank Component **T***:* Minimizing the Lagrangian with respect to **T**

$$\begin{aligned}\mathbf{T}_{k+1} &= \arg\min_{\mathbf{T}} \mathcal{L}(\mathbf{R}_{k+1}, \mathbf{N}_k, \mathbf{S}_{k+1}, \mathbf{T}, \mathbf{\Lambda}_{1,k}, \mathbf{\Lambda}_{2,k}) \\ &= \arg\min_{\mathbf{T}} \lambda \sum |\nabla \mathbf{T}| + \frac{\alpha}{2}\left\|\mathbf{T} - \left(\mathbf{R}_{k+1} - \frac{\mathbf{\Lambda}_{2,k}}{\alpha}\right)\right\|_{\text{F}}^2 \end{aligned} \tag{14}$$

yields the widely used TV denoising method [46]. Its popularity resulted in research into numerically efficient algorithms [51], [52] for solving it. These can readily be plugged into Algorithm 1.

4) Speckle Noise **N***:* The speckle noise subproblem consists of jointly applied TV and L_1 regularization terms

$$\begin{aligned}\mathbf{N}_{k+1} &= \arg\min_{\mathbf{N}} \mathcal{L}(\mathbf{B}_{k+1}, \mathbf{N}, \mathbf{S}_{k+1}, \mathbf{T}_{k+1}, \mathbf{\Lambda}_{1,k}, \mathbf{\Lambda}_{2,k}) \\ &= \arg\min_{\mathbf{N}} \mu\|\mathbf{N}\|_1 + \nu \sum |\nabla_t \mathbf{N}| \\ &\quad + \frac{\alpha}{2}\left\|\mathbf{G} + \frac{\mathbf{\Lambda}_1}{\alpha} - \mathbf{B}_{k+1} - \mathbf{N} - \mathbf{S}_{k+1}\right\|_{\text{F}}^2. \end{aligned} \tag{15}$$

Solutions for this problem are given as one of the examples in [51], or for a dedicated algorithm, the interested reader may consult [53]. As with the TV algorithms, these can be directly integrated into Algorithm 1.

It is not necessary to solve the subproblems for **N** and **T** to full convergence when running Algorithm 1. This means that, for example, the methods proposed in [51] or [52] to solve TV regularized problems can be terminated after a few iterations. Although this results in slightly off solutions, the errors are captured in the Lagrangian multipliers and addressed in later iterations. This can significantly speed up Algorithm 1 and was already mentioned for a similar minimization problem in [49].

C. Parameter Selection

Minimizing (9) requires choosing several parameters. We describe guidelines that result in sane parameter ranges, the rationale for choosing them, or provide references to direct assignments. Finally, Table I lists the parameter set used for all the experiments in this article.

For α, [34] and [54] suggest using

$$\alpha = \frac{m_1\ m_2}{4\|\mathbf{G}\|_1}$$

where m_1 and m_2 are the dimensions of the data matrix **G**. This results with near certainty in convergence of the ALM algorithm they proposed for solving (1). We observed that this is also the case for Algorithm 1.

The parameters μ and ν define how much speckle and outliers each contribute to the signal's corruption, i.e., the distribution between speckle **N** and outlier **S** energy. For avoiding that **S** absorbs all noise when minimizing (7)

$$\mu < 1$$

must hold.

The rationale for choosing ν is more elaborate. Let o be a temporal outlier and let $\mathbf{t}_-$ and $\mathbf{t}_+$ denote its adjacent temporal coordinates. For **S** to contain more of the outlier signal, the following inequality must hold:

$$|o| < \mu|o| + \nu|\mathbf{N}_{\mathbf{t}_+} - o| + \nu|o - \mathbf{N}_{\mathbf{t}_-}|$$

which is a direct consequence of the minimization terms over **N** and **S** in (7). In rough approximation, we can assume that the speckle component **N** has zero mean if it does not contain outliers, i.e., $\text{E}\left\{\mathbf{N}_{\mathbf{t}_-}\right\} = \text{E}\left\{\mathbf{N}_{\mathbf{t}_+}\right\} = 0$. On average, this turns the TV penalty, that is, a L_1 penalty on the derivative of **N**, into just another L_1 penalization on o. However, with an additional factor of two as the outlier is an extremum

$$|o| < \mu|o| + 2\nu|o|$$

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TABLE I
PARAMETERS OF (9) USED THROUGHOUT ALL EXPERIMENTS IN SECTION IV. THIS PARAMETER SET ENSURES CONVERGENCE FOR VARIOUS SAR STACKS, BOTH SYNTHETIC AND REAL DATA, REGARDLESS OF THE NUMBER OF DETAILS PRESENT AND THE GENERAL SIGNAL LEVEL

Parameter	Value	Description
w_i	$C/\left(\sqrt{\sigma_i(\mathbf{G})}+\varepsilon\right)$	weighted nuclear norm weights
C	5	constant for weights w_i
λ	0.5	low-rank TV penalty
μ	0.8	speckle L_1 penalty
ν	0.15	speckle TV penalty
α	5.0	Lagrangian multiplier update rate

which yields a lower bound for ν

$$\nu > \frac{1-\mu}{2}.$$

Zhu *et al.* [30] define the weights as

$$w_i = \frac{C}{\sqrt{\sigma_i(\mathbf{G})}+\varepsilon}$$

where ε ensures numerical stability for small singular values. Such a weight assignment penalizes small singular values more severely so that the restored signal $\mathbf{B}$ is low-rank with only a few nonzero singular values.

This leaves the parameters C and λ for tuning. Fundamentally, low rank and TV both penalize noise in $\mathbf{B}$ and changing C or λ adjusts the amount of denoising. We observed that increasing one, while appropriately decreasing the other, does not alter the level of denoising, at least within a certain range of values.

A key difference is that enforcing low-rank does not only penalize noise that cannot be well represented by a few left- and right-singular vectors of an singular value decomposition (SVD). The weighted nuclear norm also restricts signal energy as a whole, leading to generally darker output images. By introducing TV, we can to some degree get rid of this undesired behavior and still achieve the same level of denoising.

These guidelines help narrow down the search space of the possible parameters. The solving of (7) is completely agnostic to the way the similar SAR amplitude time series are collected. As such, the same parameter set should be applicable to various nonlocal filters. In Section III-D, we describe a filtering algorithm based on DespecKS [55]. After testing the proposed method on several synthetic and real data stacks, we arrived at the parameter set reported in Table I. This parameter set ensured convergence for all the tested SAR stacks, regardless of the amount of details present and the general signal level and is used throughout all experiments in Section IV.

D. Proposed Method Based on DespecKS

We briefly describe DespecKS [55], a classic nonlocal filter, such as [10], with the following filtering steps.

1) Collecting similar vectors inside a search window using the Kolmogorov–Smirnov statistical test [56].
2) Discarding similar vectors that are not spatially connected.
3) If the size of the collection does not exceed a predefined size, no filtering is performed to preserve the isolated point scatterers.
4) Computation of the nonlocal mean, without any weights, over the collection gives the filtering output.

Alike DespecKS, our algorithm also performs the similarity search with the Kolmogorov–Smirnov statistical test and, in fact, both algorithms use the same parameter set in Section IV. The subsequent steps (2–4) are where our algorithm departs from the path laid out by DespecKS.

We sort the collection of vectors by their similarities and stack them along a new axis to form the data matrix $\mathbf{G}$. Sorting imposes structure, up to a certain degree, on $\mathbf{G}$, which benefits TV regularization. In case some of the collected vectors exhibit different patterns than others, for example, by including the same temporal change, they are more likely to be grouped together.

DespecKS's nonlocal mean is replaced by (7) and the low-rank approximation's column vectors make up the denoised output at their respective spatial locations. Our algorithm does not mandate a minimum size for the vector collection, the assumption being that it is well capable of preserving the inherent structure of the collection and can consequently preserve point scatterers. In addition, although the Kolmogorov–Smirnov similarity test is susceptible to outliers, as already [55] mentioned in [55], our proposed method should be largely robust against them.

As this filtering procedure is repeated for every pixel coordinate, multiple reflectivity estimates exist for each location. We simply aggregate them by computing their unweighted mean.

This is just one possible example of a filtering algorithm. Since the computed similarities play no further role in the filtering process, relying on other statistical tests such as the ones analyzed in [57] and [58] is also possible. Different approaches to aggregate overlapping estimates can be integrated as well.

IV. EXPERIMENTS

We conducted several experiments to analyze the proposed filtering algorithm's performance and compared it with other SAR time series despeckling filters.

Because our algorithm extends DespecKS [55], we also included it in our comparison. This allows us to better judge the level of improvement provided by the proposed nonlocal low-rank approximation. Both filters use an identical set of parameters: a search window size of 21×21 and a significance level of 0.25 for the Kolmogorov–Smirnov similarity test. For DespecKS only, the minimum number of statistically homogeneous pixels for computing the average (step 3) is set to 10. As mentioned earlier, the proposed method does not require an extra treatment of point scatterers, as the low-rank approximation preserves them.

Furthermore, we compared our method with MSAR-BM3D [26] and RABASAR [28], as representatives of state-of-the-art multitemporal SAR filters. For both, we used the

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default parameter sets in their respective articles, with the exception for MSAR-BM3D, where we had to use the discrete cosine transform (DCT) instead of biorthogonal wavelets for the transform along the temporal axis in the first processing step. This was necessary to work with stack sizes that are not a power of two. We explicitly commented when this change was relevant for specific experiments.

A. Data Description

In addition to simulations, we used two SAR stacks of real data to assess all despeckling filters.

Because they are free and widely available, Sentinel-1 interferometric wide swath (IW) acquisitions with a resolution of 20 m $\times$ 5 m make up the first test set, as currently, they are probably the most common source of SAR stacks. The test set of 36 images was acquired over Mount Fuji, Japan, between February 18, 2017, and April 14, 2018, all with an incidence angle of approximately 43.5° and their VV channels were coregistered and resampled to a common master.

The second test data set consisted of a stack of 41 coregistered Sentinel-1 stripmap acquisitions' VV channel with a resolution of 5 m $\times$ 5 m, acquired between January 4, 2018, and May 11, 2019, over the island of St. Helena in the Atlantic Ocean with an incidence angle of 41.5°. Due to the data's higher resolution, man-made structures, such as roads and buildings, were visible and allowed us to analyze the ability of various methods to preserve resolution in greater detail.

Inside the comparatively small filtering windows changes in backscatter are mostly due to different land cover or singular objects (stones, buildings,…) and not the local incidence angle. Therefore, we did not perform radiometric calibration for both data sets to account for the local incidence angle using a digital elevation model.

B. Synthetic Data

The first experiment illustrates the signal's decomposition into its low-rank, sparse outlier, and speckle components. We construct a matrix $\in \mathbb{R}^{32\times 100}$, the columns of which are the absolute value of independent samples of a multivariate, circular complex Gaussian random variable [59], with mean 1 and exponential coherence loss with $\lambda = 0.3$. Each column contains the correlated Rayleigh distributed values, simulating a speckle corrupted SAR amplitude stack. We then replace 1% of the matrix's entries by Ricean distributed outliers. The parameters for the probability density function

$$p_{\sigma,\nu}(x) = \frac{x}{\sigma^2} e^{-\frac{x^2+\nu^2}{2\sigma^2}} I_0\left(\frac{x\nu}{\sigma^2}\right)$$

where I_0 is the modified Bessel function of the first kind with order zero, $\sigma = 1$ and $\nu = 4$.

To imitate the search for similar vectors within a search window, we randomly selected 40 column vectors. Stacking these vectors along a new axis forms the noisy data matrix $\mathbf{G}$. After applying (7), the final result was obtained by aggregating the low-rank component's column vectors at their original positions with previous estimates. This process was repeated 60 times, mimicking the joint denoising inside a search window.

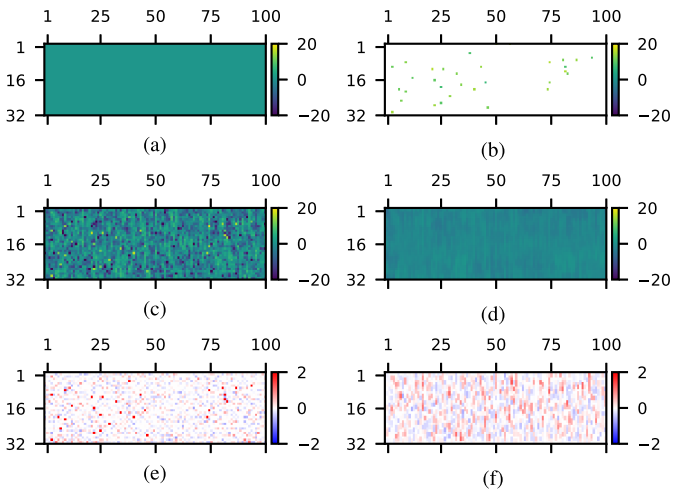


Fig. 3. Synthetic experiment to analyze the decomposition of the noisy signal (c) into its low-rank approximation (d), estimated outliers (e), and speckle components (f). (a) and (b) True signal and outliers. (a)–(d) Plotted in dB.

Fig. 3 shows the resulting components and the original signals. The low-rank approximation mostly resembles the original, true signal. More interesting are the estimated outliers and the speckle noise. Visible vertical stripes in the speckle noise reflect the temporarily correlated speckle component, its uncorrelated parts, due to the loss of coherence, which are absorbed in the estimated outlier image and are visible as small perturbations around zero. By comparing the outlier image with the true outliers, we can see that they are reasonably well extracted from the noisy input signal.

Experiments on a more realistic, simulated SAR stack, corrupted by varying percentages of outliers, provide a quantitative analysis and give an impression on each methods' robustness to outliers. We compare all methods in terms of peak signal-to-noise ratio (PSNR), mean structural similarity (MSSIM) index [60], equivalent number of looks (ENL) [61], and error vector magnitude (EVM) [62]. We assume that the reader is already familiar with PSNR, MSSIM, and ENL and have not explicitly covered them here; however, we have briefly introduced EVM.

EVM compares the error vector's root mean square P_e to the reference signal's root mean square amplitude P_r

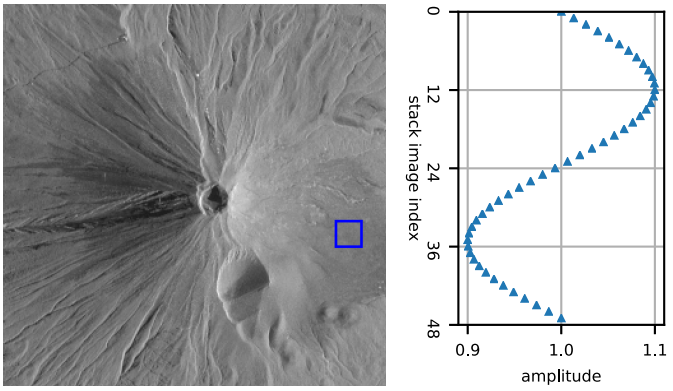
$$\mathrm{EVM}_{\mathrm{dB}} := 10\log_{10}\frac{P_e}{P_r}$$

with lower values of EVM indicating better performance.

In our analysis, we extracted all pixel values at outlier coordinates and compare them to the uncorrupted stack by computing the respective EVM.

For generating a synthetic SAR time series, we used the previously described Sentinel-1 IW stack of Mount Fuji to create an essentially speckle-free single amplitude image. Starting from the coregistered input, spatial multilooking by a factor of 8 in range and 2 in azimuth results in speckle reduction and pixels with an essentially quadratic footprint.

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(a) (b)

Fig. 4. Input data used for generating a synthetic SAR stack. Temporal mean amplitude image (a) 36 image Sentinel-1 IW stack with additional spatial multilooking (8 in range, 2 in azimuth). This image is multiplied by the amplitude profile in (b) generate a stack with slight temporal changes.

TABLE II
PSNR, MSSIM, AND EVM VALUES FOR DIFFERENT PERCENTAGES OF OUTLIERS

	%	DespecKS	MSAR-BM3D		RABASAR	DespecKS-NLLRTV
			ID	DCT		
PSNR in dB	0	28.519	**29.964**	29.606	29.398	27.279
	0.1	28.589	28.571	27.546	**29.387**	27.293
	0.5	28.797	26.681	26.487	**29.267**	27.343
	1.0	28.848	25.798	25.818	**29.086**	27.402
MSSIM	0	0.672	0.653	**0.720**	0.685	0.694
	0.1	0.665	0.535	0.438	0.677	**0.694**
	0.5	0.641	0.328	0.314	0.630	**0.693**
	2.0	0.618	0.254	0.255	0.579	**0.692**
ENL	1	26.521	**105.647**	13.289	99.834	35.884
	0.1	25.878	26.285	4.071	**88.343**	35.673
	0.5	17.406	0.135	1.243	**74.723**	35.869
	1.0	30.353	0.077	0.267	14.945	**37.019**
EVM	0.1	-3.304	6.098	4.154	**-6.994**	-6.387
	0.5	-4.056	7.592	5.270	-3.665	**-6.432**
	1.0	-4.107	7.885	6.243	-2.410	**-6.457**

Computing the temporal mean further eliminates any traces of speckle. Fig. 4(a) shows the final 300 × 300 pixel image used in our simulation. The blue rectangle delineates a homogeneous area that we used for computing the ENL.

From this image, we generated an uncorrupted and noise-free SAR stack by computing its outer product with the amplitude profile shown in Fig. 4(b), imitating slight temporal changes between acquisitions.

The correlated speckle is added using multivariate, circular complex Gaussian random variables and exponential coherence decay with $\lambda = 0.3$, identical to the previous experiment. This stack is then further corrupted by replacing various percentages of pixels with Ricean distributed outliers.

Table II lists PSNR, MSSIM computed on the stack in dB, ENL, and the EVM computed on all outlier positions. There are several characteristics to note. The performance generally degrades with more outliers, except for the DespecKS-based algorithms. We assume this is because of the search for similar vectors, which becomes less discriminated if the pixels are corrupted by outliers.

The performance of MSAR-BM3D with DCT as a temporal transform in its first step is misleading, since this transform perfectly captures the amplitude profile [Fig. 4(b)], leading to a very sparse and therefore easy-to-threshold representation. For a fairer comparison, we conducted the same experiment but replaced DCT with no transform, which for the no outlier case drastically worsened the MSSIM index (Table II ID column).

It should be noted that for high percentages of outliers, MSAR-BM3D breaks down and neither speckle nor outliers are removed. This manifests as ENL values that are smaller than one and is also indicated by the low MSSIM and high EVM values. Thus, the area can no longer be considered as homogeneous when computing the ENL.

The generally poor results of our proposed method in terms of PSNR are due to its penalization of signal energy, which we already mentioned in Section III-C when discussing the interplay of low-rank and TV regularization. Its capacity to preserve details, indicated by MSSIM, is, however, unsurpassed. In addition, as particularly evident by the EVM values, it is virtually unaffected by outliers.

For a visual interpretation of the numbers in Table II, Fig. 5 shows the results for the stack's 30th image in the top row with no outliers and in the bottom row if 0.5% of all the pixels were corrupted by outliers.

With no outliers present, MSAR-BM3D unarguably shows the best performance, corroborating its result in Table II. However, as already mentioned, the amplitude profile is a perfect match for the employed DCT and gives MSAR-BM3D, in this case, an unfair advantage. In addition, if outliers corrupt the data, its filtering performance is considerably hamstrung. In comparison with Fig. 4(a), it becomes clear that RABASAR tends to produce oversmoothed estimates, yet visibly, its despeckling performance is largely unaffected by outliers. The original DespecKS algorithm, while being largely robust against outliers, erases some fine details as the road in the top left corner. As already indicated by its MSSIM and EVM values, our proposed method is virtually unaffected by outliers and also preserves fine details. However, due to the penalization of signal energy in the low-rank component, it is unable to preserve radiometric accuracy, and its filtering results are slightly darker.

Generally speaking, neither MSAR-BM3D nor RABASAR remove all outliers, especially in the areas of low radar reflectivity. Admittedly, outliers could also be considered as a part of the signal, and their preservation is desired.

C. Real Data

We first present the results for the Sentinel-1 IW stack over Mount Fuji, Japan. Fig. 6 shows the various filters' results for the acquisitions on March 26, 2017, April 7, 2017, and April 19, 2017, of parts of Mount Fuji. These images are single look and therefore skewed due to the different range and azimuth resolutions (5 and 20 m) of IW acquisitions. They show in their top left corner, the peak of Mount Fuji; and in

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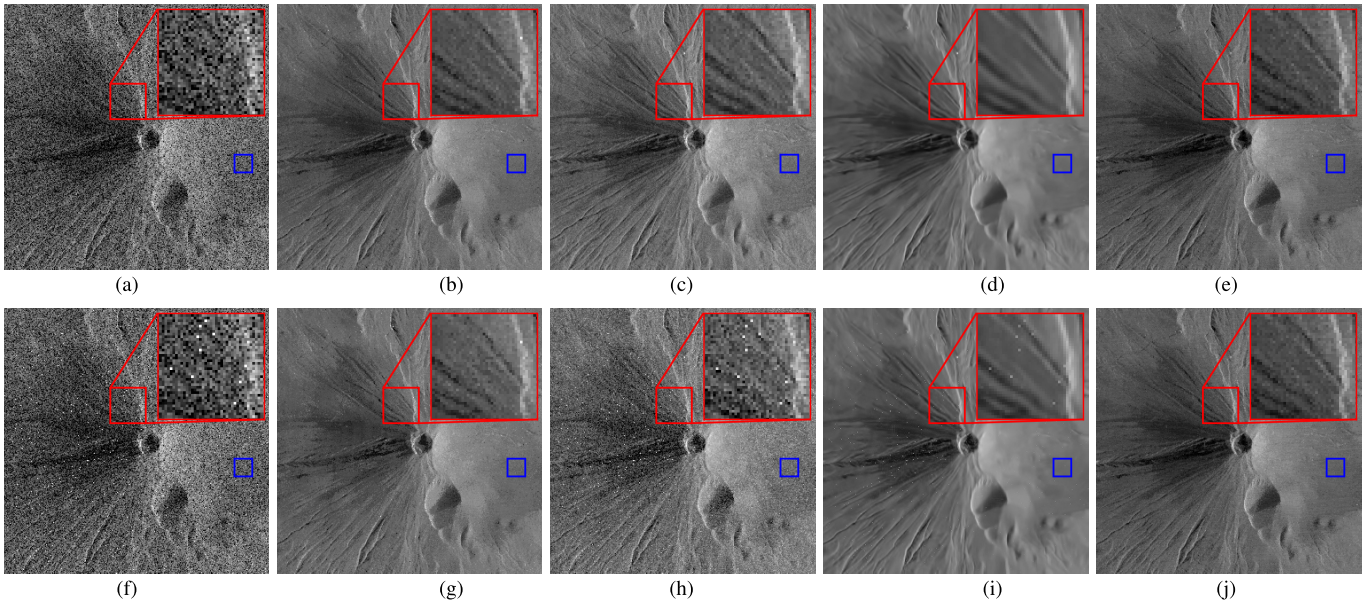


Fig. 5. Filtering results for the synthetic stack of 48 images introduced in Fig. 4. With no outliers (a) top row and with 0.5% corrupted pixels (b) in the bottom row. (b) and (g) DespecKS, (c) and (h) MSAR-BM3D DCT, (d) and (i) RABASAR, and (e) and (j) DespecKS-NLLRTTV. Both MSAR-BM3D and RABASAR do not remove outliers.

the center, the Hoeizan crater. The most striking difference between the filters is their treatment of the outlier artifact at Mount Fuji's Hoeizan crater in the acquisition of April 7. This artifact is present only in one of the 36 acquisitions, and while our proposed method detects and removes it, other filters preserve it. Another noteworthy characteristic of our proposed method is that it lessens the magnitude of temporal deviations. This is visible around Mount Fuji's crater for the acquisition on April 7, where the radar backscatter is adjusted to more closely match the levels of other acquisitions. Whether this behavior is desired or not depends on the use case and the kind of further analysis performed on the data. Regardless, permanent changes will be preserved, as they will not be considered as sparse outliers nor be penalized by TV regularization on the low-rank component.

Regardless of this characteristic, we can make the following observations about the filters' despeckling performances. DespecKS oversmoothens the finer details and its estimate contains some single-pixel outliers. Both deficiencies are not present in DespecKS-NLLRTV's output, even though it uses exactly the same methodology and parameters to search for similar pixels. MSAR-BM3D offers noise suppression and detail preservation that are comparable to the proposed method, but additionally it does not lead to erroneous darker images. RABASAR preserves details but introduces a grain effect in homogeneous areas.

Table III shows ENL values calculated separately for each acquisition for the zoomed-in areas in Fig. 6. The results are mostly consistent with the simulation results reported in Table II, i.e., combining nonlocal low-rank TV regularization with DespecKS improves performance. DespecKS-NLLRTV's ENL values vary the least between acquisitions due to its regularization along the temporal axis.

TABLE III
ENL VALUES FOR THE ZOOMED-IN AREAS OF THE THREE ACQUISITIONS OF FIG. 6

Date	DespecKS	MSAR-BM3D DCT	RABASAR	DespecKS-NLLRTV
March 26th	57.758	58.570	36.015	**120.197**
April 7th	55.817	37.496	18.407	**113.296**
April 19th	78.205	76.927	19.389	**110.274**

At last, Fig. 7 shows filtering results of the Sentinel-1 stripmap stack acquired over St. Helena. In particular, it depicts St. Helena's newly constructed airport on May 4, 2018. At the terminal, vehicles visible as bright point scatterers move or disappear between acquisitions, leading to artifacts in the filtering results. By comparing the filtering results to the unfiltered data and the temporal mean, we can determine to what extent a specific filtering algorithm can cope with temporal changes. Since DespecKS is a purely spatial filter, it does not introduce temporal artifacts, but it tends to smooth out bright point scatterers. RABASAR offers good noise reduction but, as in previous experiments, tends to oversmoothen the image and cannot deal with all temporal changes appropriately. As in the previous example, MSAR-BM3D and DespecKS-NLLRTV provide comparable levels of detail preservation and noise reduction. How they handle outliers is where they differ. Fig. 7(h) shows the outlier image produced by DespecKS-NLLRTV. It captures not only some uncorrelated speckle but also one of the moving vehicles, which is then removed from the filtered output.

Furthermore, a direct comparison of DespecKS and DespecKS-NLLRTV shows the benefit of replacing a simple mean with the more advanced low-rank approximation

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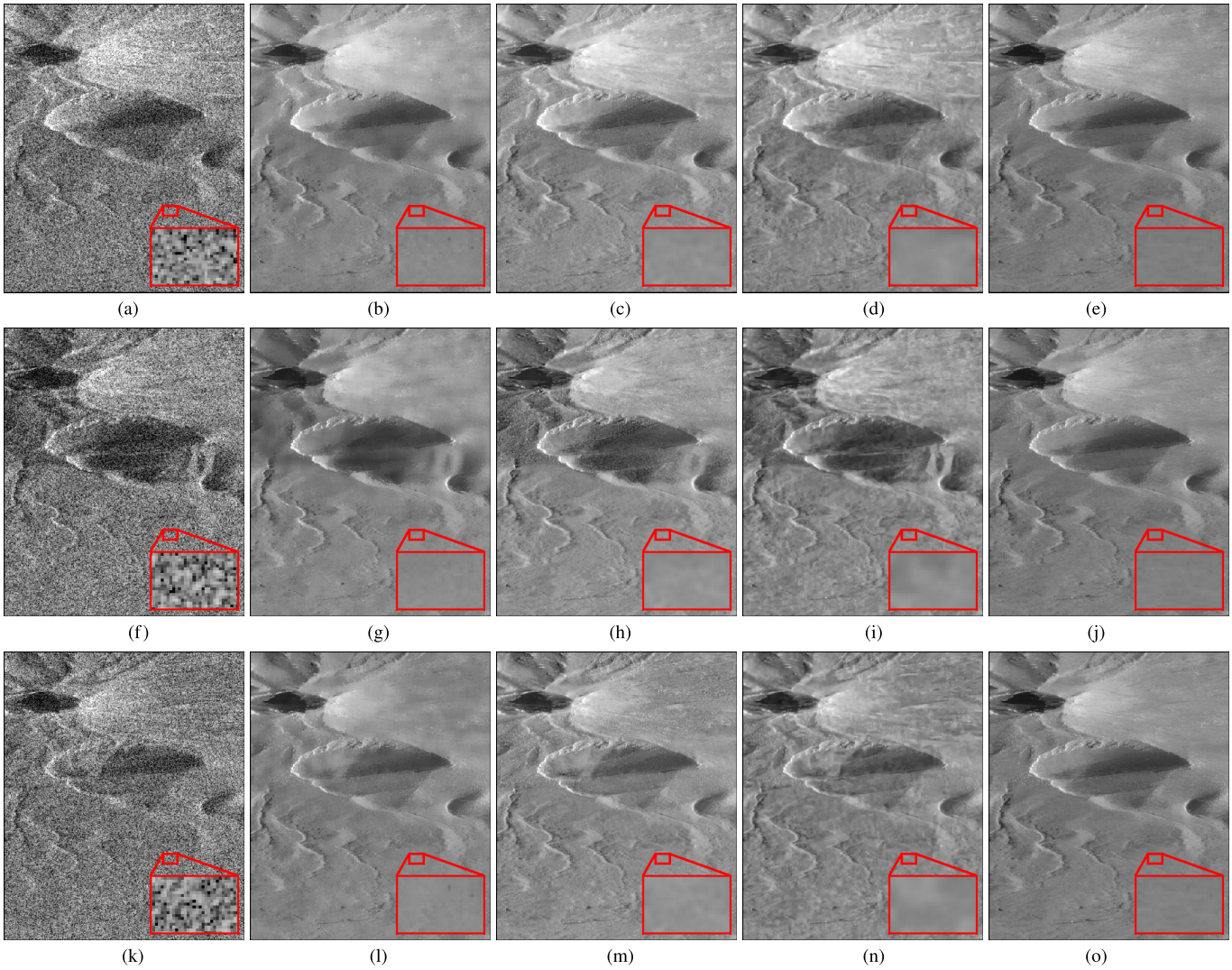


Fig. 6. Denoising results for three consecutive out of 36 Sentinel-1 IW acquisitions. The images show in the top left corner the peak of Mount Fuji and in the center the Hoeizan crater. (a), (f), and (k) Unfiltered data from March 26th, April 7th, and April 19th. (b), (g), and (l) DespecKS. (c), (h), and (m) MSAR-BM3D. (d), (i), and (n) RABASAR. (e), (j), and (o) DespecKS-NLLRTV. The data acquired April 7th (f) contains an outlier artifact at the crater not present in all the other acquisitions.

together with TV regularization, which leads to better preservation of point targets.

Again, we point out that DespecKS-NLLRTV's output is slightly darker than that of others due to the low-rank penalization of absolute signal energy.

V. Discussion

The proposed nonlocal low-rank model for SAR time series can serve as a drop-in replacement for existing nonlocal means stack despeckling filters. Section IV showed that extending DespecKS with low-rank approximation resulted in enhanced detail preservation by directly including the temporal domain and also increased robustness against outliers. The resulting filter proved to be competitive with other state-of-the-art multitemporal despeckling filters.

The separation of outliers merits a dedicated discussion. Section IV showed that the proposed approach can tackle very different kinds of outliers. The single pixel perturbations as shown in Fig. 5, the large artifact in Fig. 6, or one of the randomly appearing and disappearing airport vehicles in Fig. 7 are all identified as outliers. An outlier's nature and the application the user is interested in determine whether it should be considered as noise or an interesting anomaly. This requires a certain level of understanding to properly apply the proposed method, because the outlier signal component might actually include interesting information, as, for example, in Fig. 7. However, this separation of outliers from speckle in the proposed method makes these anomalies easier to detect and analyze. The desired outlier component can be added to the low-rank component, possibly after thresholding to remove low-amplitude noise as shown in Fig. 7(h). To a lesser degree, such a level of understanding is already necessary for existing speckle filters, where all users are aware that they might end up compromising spatial details.

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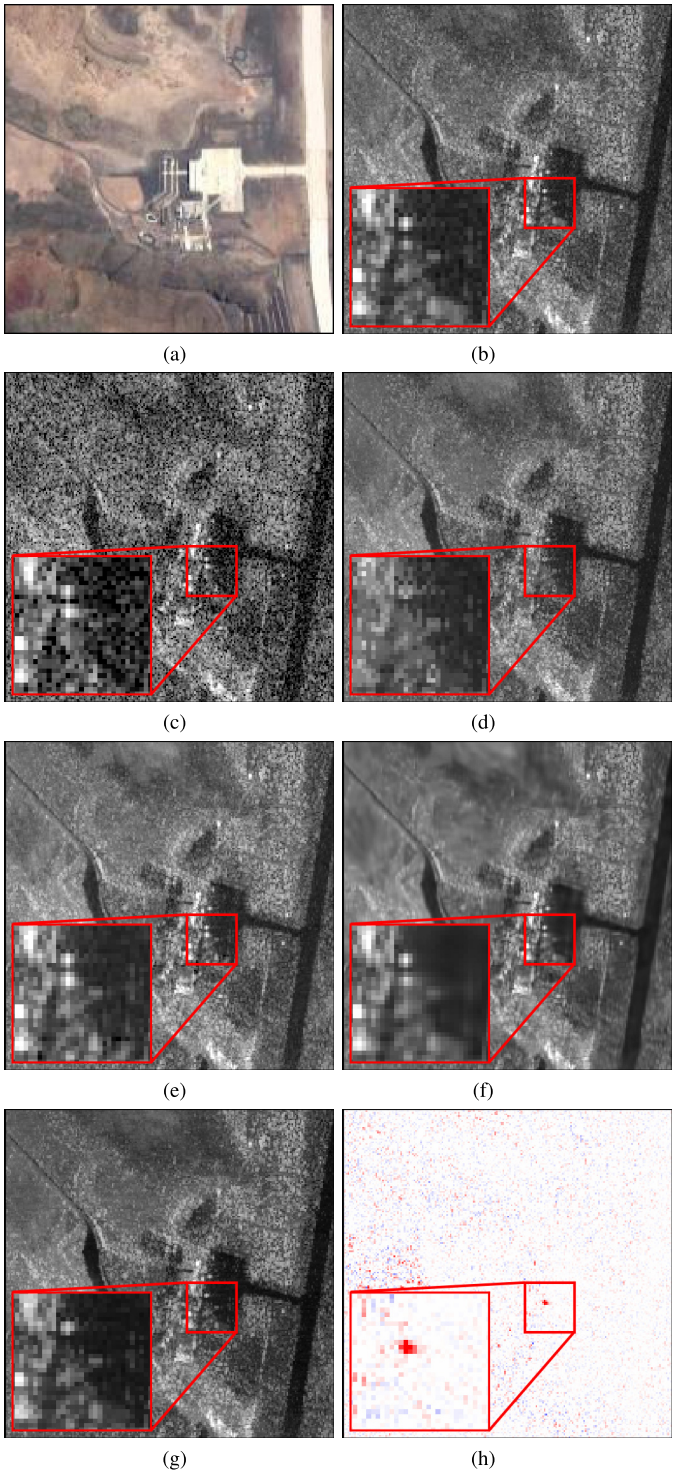


Fig. 7. St. Helena's airport acquired on May 4, 2018. Vehicles, visible as bright point scatterers, move between acquisitions, leading to filtering artifacts. (a) Google maps. (b) Mean. (c) Unfiltered data. (d) DespecKS. (e) MSAR-BM3D. (f) RABASAR. (g) DespecKS-NLLRTV. (h) DespecKS-NLLRTV outliers.

Another interesting point is the low-rank model [see (7)] itself. The most relevant parts are the weighted nuclear norm and the L_1 penalty. The fact that such a comparatively simple model is applicable to SAR data is somewhat surprising. SAR, with its signal-dependent noise, usually requires dedicated algorithms and approaches for despeckling, whereas the weighted nuclear norm and L_1 penalty are quite straightforward. One obvious drawback is the model's inability to preserve radiometric signatures, due to the penalization of energy in the nuclear norm. This might not pose a problem if only geometrical features or relative changes are of interest. Especially contemporary machine learning algorithms can be made robust against distortions such as changes in the overall brightness or geometric transformations.

Thus far, we have ignored the computational costs of the proposed method. In conventional nonlocal means filters, the computation of patch similarities dominates the overall costs due to the computational simplicity of a weighted mean. This ratio is reversed for our approach. Examining Algorithm 1 more closely shows that the operations within the loop take up most computational resources. The computational costs of soft thresholding are negligible and both TV minimization problems need not be solved to full accuracy [49], as errors are handled by the Lagrangian multipliers $\mathbf{\Lambda}_1$ and $\mathbf{\Lambda}_2$ in later iterations. Thus, the SVD incurs the greatest costs. The computation cost can be lowered by replacing the exact SVD with an approximate approach such as randomized SVD [63]. Furthermore, because the minimization has to be performed for every spatial coordinate of the stack, batched SVD routines can speed up their computations [64]. Even with such modifications, we estimate that our approach is still considerably slower than, for example, MSAR-BM3D or RABASAR.

Finally, although imposing TV regularization in the temporal direction on the speckle component results in the desired correlation, it is a static, global constraint. In real SAR data, coherence changes with land cover and time of the year. A more flexible mechanism should yield better results for extracting outliers from speckle.

We see this study's approach only as a first step that entails many possibilities for improvement.

VI. Conclusion and Future Work

We proposed a low-rank approximation model for SAR time series. Integrating it into the well-known DespecKS algorithm resulted in a filter with greater robustness against outliers and mismatches in the similarity search. In addition, the proposed method can, up to a certain degree, separate singular outliers from speckle by taking the correlation of speckle between SAR acquisitions into account.

Interesting applications of SAR time series were already alluded to in the Introduction. Due to their all-weather capability, SAR stacks provide regular updates, not inhibited by cloud cover, as opposed to optical sensors. By essentially removing all speckle, modern despeckling filters, such as MSAR-BM3D, RABASAR, or our proposed method, significantly ease any subsequent analysis. With such a clean data set, even researchers unfamiliar to SAR might start using time series delivered by current and future missions in their respective fields.

In future research, we wish to extend the low-rank approach to tensors, by going from single pixels to spatial patches, including multiple polarizations, incidence angles, and maybe ascending and descending acquisition orbits into the estimate. Methods that deal with missing data and Gaussian noise

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already exist [65]; however, they have not been adapted to SAR. Further research is also needed to speed up the optimization problems, which has limited the large-scale application of our method. This can be either achieved through more sophisticated optimization algorithms than Algorithm 1 or a more efficient implementation, possibly by using batched basic linear algebra subprograms (BLAS) implementations, which are becoming increasingly relevant.

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